Safety Zone Characterization in Elderly Hypertensive Patients Using Cardiopulmonary Exercise Testing:

A 2×2 HTN×EIH Phenotypic and Machine Learning Classification Study (N=3,206)

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Abstract

Background: Cardiopulmonary exercise testing (CPET) is the gold standard for assessing exercise safety in cardiovascular patients, yet a systematic safety zone stratification framework for elderly hypertensive patients remains lacking.

Objectives: (1) Characterize CPET phenotypes across hypertension history (HTN) and exercise-induced hypoxemia (EIH) strata using a 2×2 factorial design; (2) Develop a dual-timeline safety zone prediction framework (pre-exercise P0 + post-exercise P1); (3) Quantify the predictive ceiling of summary-level CPET data with explicit Leakage Guard protection.

Methods: Retrospective single-center study (N=3,206) using a 2×2 design: HTN history × EIH response. Four cohorts: CTRL (n=1,858), HTN-NoEIH (n=738), HTN-EIH (n=277), EIH-Only (n=333). A 5-layer quality control pipeline with explicit Leakage Guard protection was implemented. Type II two-way ANOVA (partial η^2), Dunn's post-hoc comparisons, reference equations (OLS, n≈968), and Logistic regression for EIH predictors characterized the phenotypic structure. Machine learning: P0 (LASSO + XGBoost); P1 (Ordinal Logistic + LightGBM + CatBoost); SHAP global feature importance.

Results: HTN history was the dominant driver of reduced aerobic capacity (VO_2peak F=290.6, p<0.001, partial η^2=0.084, medium effect); EIH response primarily manifested as markedly elevated peak blood pressure (F=1803.1, p<0.001, η^2=0.371, large effect). The HTN×EIH interaction on VO_2peak was non-significant (F=0.02,

p=0.894), indicating independent rather than synergistic effects. HTN groups showed median VO_2peak 19.2–20.1 vs 23.2–23.5 mL/kg/min in non-HTN groups. Safety zone distribution: Green 40.2%, Yellow 34.3%, Red 24.9%; HTN-EIH Red zone 99.3%, CTRL Red zone 6.3%. P0: XGBoost AUC=0.583; P1: CatBoost $F1_{macro}$=0.499, κ=0.302; reflecting summary-level predictive ceiling. SHAP identified VO_2peak and peak heart rate as dominant features.

Conclusions: HTN and EIH affect distinct cardiopulmonary dimensions through independent main effects rather than synergistic interaction. Summary-level CPET data approaches its predictive ceiling (P0 AUC≈0.58, P1 F1≈0.50), necessitating breath-by-breath waveform data for further improvement. The Leakage Guard methodology quantified the leakage effect at ≈0.37 F1 units, providing a reproducible framework for retrospective CPET studies.

Keywords: Cardiopulmonary exercise testing; Exercise safety; Hypertension; Exercise-induced hypoxemia; Machine learning; Safety zone

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1 Introduction

Cardiovascular disease (CVD) is the leading cause of mortality and disability in elderly populations worldwide [1]. Exercise-based cardiac rehabilitation significantly improves cardiopulmonary function and prognosis [2, 3], yet exercise-induced adverse events—particularly exercise-induced hypoxemia (EIH) and hypertensive stress responses—remain significant barriers to safe exercise prescription [4, 5].

Cardiopulmonary exercise testing (CPET) provides a comprehensive assessment of cardiorespiratory reserve and is the gold standard for exercise safety evaluation [6, 7]. However, a systematic CPET-based safety zone classification for elderly hypertensive patients remains underdeveloped: (1) the interaction between hypertension history and EIH has not been systematically quantified in large cohorts; (2) the predictive ceiling of summary-level CPET data for machine learning is unknown; (3) data leakage—a pervasive methodological issue in retrospective CPET modeling—is rarely explicitly addressed [8].

Machine learning applied to CPET data has yielded promising results: Li et al. [9] achieved AUC=0.794 in coronary artery disease patients, Arina et al. [10] reached AUC=0.946 in perioperative elderly patients, but both included extensive clinical laboratory features (medications, echocardiographic data) [11, 12]. The Chinese CPET expert consensus [13] recommends standardized CPET protocols, but quantitative safety zone frameworks for elderly hypertensive patients based on large-scale data remain lacking.

This study aims to: (1) characterize CPET phenotypes using a HTN×EIH 2×2 design; (2) develop a dual-timeline safety zone framework (P0 pre-exercise + P1 post-exercise); (3) establish Leakage Guard methodology to quantify and prevent data leakage in retrospective CPET modeling.

2 Methods

2.1 Study Population and Design

This retrospective single-center cohort study enrolled patients ≥60 years who completed CPET at our institution from 2015 to 2023. Of 3,232 eligible records, 3,206 passed quality control (26 excluded, 0.8%) and were included in analysis.

A 2×2 factorial design stratified patients by:

- Factor A: HTN history (prior diagnosis of hypertension: Yes/No)
- Factor B: EIH response (exercise-induced SpO_2 drop ≥4% or nadir <90%: Yes/No)

Table 1: HTN×EIH 2×2 Cohort Design

Cohort	Code	N (post-QC)	Description
Control/non-HTN	CTRL	1,858	No HTN history, no EIH
HTN history, no EIH	HTN-NoEIH	738	HTN history, no EIH during test
HTN history + EIH	HTN-EIH	277	HTN history + EIH during test
EIH only	EIH-Only	333	No HTN history, EIH during test
Total		3,206	(3,232 pre-QC)

This study was approved by the Institutional Review Board (Approval No.: to be filled). Data were fully de-identified.

2.2 CPET Protocol and Variable Definitions

CPET was performed using an incremental cycling or treadmill protocol with a standard metabolic gas analysis system. Key variables: VO_2peak (mL/kg/min), HR_{peak} (bpm), O_2 pulse (mL/beat), VT1-VO_2, VE/VCO_2 slope, heart rate recovery at 1 minute (HRR1), oxygen uptake efficiency slope (OUES), peak METs.

EIH definition: SpO_2 drop $\geq$4% from rest or nadir $<$90% during exercise. In this dataset, EIH was identified from the group_code field (groups containing "EHT").

2.3 Quality Control Pipeline

A 5-layer QC pipeline was applied: (1) physiological range compliance (35 outlier values clipped); (2) cross-field logical consistency; (3) missing rate threshold check ($>$30% flagged; medication fields entirely absent); (4) protocol consistency standardization (cycling/treadmill); (5) EHT_ONLY group-specific diagnostics (139 missing peak SBP values).

Leakage Guard: Since eih_status directly participates in P1 zone definition (P1 Red criterion), and ve_vco2_slope and vo2_peak_pct_pred participate in P1 label construction, an explicit Leakage Guard mechanism was established. All label-participating fields were forcibly excluded from model features. This prevented inflated model performance: leaked model F1=0.84 vs. clean model F1=0.47, a leakage effect of $\approx$0.37 F1 units [8, 14].

2.4 Safety Zone Definitions

A dual-timeline design was adopted:

P0 (pre-exercise prediction of exercise-induced adverse events, binary): P0 labels are defined by exercise-derived measurements, while the P0 model uses only pre-exercise baseline information (age, sex, BMI, medical history, etc.) to predict whether adverse events will occur during or after exercise. This constitutes a prospective prediction task despite the retrospective study design, with strict separation between feature time-point (pre-exercise) and label time-point (during/post-exercise). Positive if any criterion is met: (1) eih_status = True (SpO_2 drop $\geq$4% or absolute <90% during exercise); (2) VO_2peak <50% predicted (post-exercise); (3) Peak SBP >220 mmHg (during exercise).

P1 (post-exercise safety zone stratification, 3-class ordinal): P1 is defined using three exercise-derived variables and employs a take-worst strategy (most severe zone assigned when criteria conflict):

- **Green (low risk)**: eih_status = False, AND VO_2peak $\geq$70% predicted, AND VE/VCO_2 slope $\leq$30
- **Yellow (moderate risk)**: eih_status = False, AND (VO_2peak 50–70% OR VE/VCO_2 slope 30–36; not meeting Green/Red criteria)
- **Red (high risk)**: eih_status = True (independent sufficient condition), OR VO_2peak <50% predicted, OR VE/VCO_2 slope >36

Note (Label Construction Axiom): EIH positivity (eih_status = True) is an independent sufficient condition for Red zone assignment — EIH+ patients are axiomatically classified as Red regardless of VO_2peak or VE/VCO_2 values. This reflects clinical consensus that EIH constitutes a high-risk marker [13]. Consequently, the near-100% Red proportion in EIH+ cohorts is a structural consequence of the label definition, not a model prediction. The Leakage Guard mandatorily excludes eih_status from P1 model features, ensuring the model cannot exploit this axiomatic constraint; P1 model predictions are meaningful for non-EIH patients.

2.5 Statistical Analysis

Baseline characteristics: Kruskal-Wallis test (non-normal continuous variables) and χ^2 test (categorical variables); Dunn's post-hoc pairwise comparisons with Bonferroni correction (α=0.0083 for 6 pairs).

HTN×EIH interaction: Type II two-way ANOVA with partial η^2 (negligible <0.01, small 0.01–0.06, medium 0.06–0.14, large $\geq$0.14, per Cohen 1988) [15]. Given the large sample size (N=3,206), even negligible effects ($\eta^2 < 0.01$) can reach statistical significance.

Effect sizes (η^2) rather than p-values serve as the primary interpretation criterion; effect size categories are indicated in table footnotes ($^n/^s/^m/^L$).

Reference equations: OLS regression on reference-normal subset with age, sex, and age×sex interaction as predictors, outputting %predicted and z-scores.

EIH predictors: Univariable and multivariable Logistic regression.

2.6 Machine Learning Models

P0 models: LASSO logistic regression (baseline, L1 regularization) and XGBoost (primary model). 5-fold cross-validation with isotonic calibration. Metrics: AUC-ROC, AUPRC, Brier score.

P1 models: Ordinal logistic (baseline), LightGBM (primary), CatBoost (comparison). Ordinal 3-class output (green=0/yellow=1/red=2), class-balanced training. Metrics: $F1_{macro}$, Cohen's κ, per-class recall.

Cost-sensitive sensitivity analysis: LightGBM with Red class weight×4, quantifying the F1-vs-safety trade-off [16].

Feature importance: SHAP (SHapley Additive exPlanations) [14].

3 Results

3.1 Baseline Characteristics

Baseline characteristics are presented in Table 2. The median age across all cohorts was 67 years with no significant difference (p=0.375). HTN groups (HTN-NoEIH and HTN-EIH) had significantly higher body weight than non-HTN groups (p<0.001), consistent with metabolic comorbidities in hypertension. HTN-NoEIH showed the lowest median VO_2peak (19.2 mL/kg/min), while EIH-Only showed the highest (23.5 mL/kg/min). HTN-EIH had the highest median peak SBP (214 mmHg). Post-hoc analysis (Dunn's, Bonferroni corrected) identified significant pairwise differences in 10 of 11 key variables.

Table 2: Baseline characteristics by 2×2 cohort (Table 1)

Variable	CTRL (n=1,858)	HTN-NoEIH (n=738)	HTN-EIH (n=277)	EIH-Only (n=333)	*P*
Continuous variables (median [Q1,Q3])					
Age (years)	67 [63,72]	67 [63,71]	67 [63,71]	67 [64,71]	0.375
VO_2peak (mL/kg/min)	23.2 [19.5,28.3]	19.2 [16.2,23.0]	20.1 [16.8,23.4]	23.5 [19.3,28.9]	<0.001
VO_2peak %pred (%)	77 [66,88]	72 [62,86]	80 [67,91]	79 [68.5,92]	<0.001
HR_{peak} (bpm)	154 [140,167]	134 [119,148]	145 [129,158]	158 [145,170]	<0.001
Peak SBP (mmHg)	161 [144,178]	171 [155,184]	214 [204,226]	212 [200,223]	<0.001
VE/VCO_2 slope	26.7 [24.2,29.4]	27.8 [25.3,30.8]	27.6 [24.8,30.0]	27.4 [24.4,30.0]	<0.001
O_2 pulse (mL/beat)	10.0 [8.0,13.0]	10.0 [8.2,13.0]	10.0 [8.0,13.0]	11.0 [8.0,14.0]	0.036
Categorical variables (n, %)					
Female	926 (49.8%)	312 (42.3%)	135 (48.9%)	147 (44.1%)	0.004
HTN history	0 (0%)	738 (100%)	277 (100%)	0 (0%)	<0.001
EIH (exercise HTN)	0 (0%)	0 (0%)	277 (100%)	333 (100%)	<0.001

Continuous: Kruskal-Wallis; categorical: χ^2. Dunn's post-hoc: 10/11 variables with significant pairwise differences. Diabetes and comorbidity fields were entirely absent in the current dataset.

3.2 HTN×EIH Interaction Effects (Two-way ANOVA)

Results of Type II two-way ANOVA are presented in Table 3.

Table 3: HTN×EIH two-way ANOVA results (Table 2)

Outcome	HTN main effect			EIH main effect			Interaction (A×B)	
	F	*p*	η^2	*F*	*p*	η^2	*F* (*p*)	η^2
VO_2peak	290.6	<0.001	0.084^m	4.37	0.037	0.001^n	0.02 (0.894)	0.000^n
VO_2peak %pred	22.3	<0.001	0.007^n	25.9	<0.001	0.008^n	5.03 (0.025)	0.002^n
HR_{peak}	487.6	<0.001	0.133^m	42.4	<0.001	0.013^s	12.87 (<0.001)	0.004^n
Peak SBP	65.6	<0.001	0.021^s	1803.1	<0.001	0.371^L	5.20 (0.023)	0.002^n
VE/VCO_2 slope	38.2	<0.001	0.012^s	0.34	0.558	0.000^n	3.37 (0.067)	0.001^n
O_2 pulse	2.91	0.088	0.001^n	9.67	0.002	0.003^n	5.84 (0.016)	0.002^n

Effect size: ^{n} negligible (<0.01), ^{s} small (0.01–0.06), ^{m} medium (0.06–0.14), ^{L} large (>0.14) (Cohen 1988).

Key findings (Figure 1): HTN history significantly reduced VO_2peak (η^2=0.084, medium effect) and HR_{peak} (η^2=0.133, medium effect). EIH response primarily drove a large increase in peak SBP (η^2=0.371, large effect; cell means: EIH+ 213 vs. EIH− 163 mmHg),

with negligible independent effect on VO_2peak. The HTN×EIH interaction on VO_2peak was non-significant (F=0.02, p=0.894, η^2=0.000). Although some interaction effects reached statistical significance in this large sample (VO_2peak%pred: p=0.025; HR_{peak}: p<0.001; peak SBP: p=0.023), all interaction η^2 values were negligible (≤0.004), indicating no clinically meaningful synergistic interaction between HTN and EIH. The two factors operate through independent main effects on distinct cardiopulmonary dimensions.

3.3 Reference Equations

The reference-normal subset (n≈968, 479 male, 489 female) yielded reference equations for five key CPET variables (Table 4).

Table 4: Reference normal equations (Table 3)

Variable	N	R^2	Key coefficients
VO_2peak (mL/kg/min)	968	0.298	Intercept=29.38; age coeff=−0.092; sex×age interaction
VO_2peak %pred (%)	969	0.013	Limited predictability from age/sex
HR_{peak} (bpm)	966	0.082	Intercept=134.5; age coeff=+0.257
VE/VCO_2 slope	969	0.040	Intercept=30.5; age coeff=−0.067
O_2 pulse (mL/beat)	968	0.489	Intercept=11.37; age coeff=−0.032

All models include age, sex (M/F), and age×sex interaction. VO_2peak R^2=0.298 indicates substantial individual heterogeneity in the elderly population (residual SD=6.3 mL/kg/min). O_2 pulse R^2=0.489 reflects the strong body-size dependence of stroke volume.

3.4 Safety Zone Distribution

Overall zone distribution (Figure 2): Green 1,289 (40.2%), Yellow 1,099 (34.3%), Red 797 (24.9%), missing 21 (0.7%).

Distribution by 2×2 cohort:

- HTN-EIH (n=277): Red 275 (**99.3%**), Green/Yellow 0
- EIH-Only (n=333): Red 324 (**97.3%**), Green/Yellow 0
- HTN-NoEIH (n=738): Green 299 (40.5%), Yellow 353 (47.8%), Red 81 (11.0%)
- CTRL (n=1,858): Green 990 (53.3%), Yellow 746 (40.2%), Red 117 (**6.3%**)

P0 event rate: 727/3,206 (22.7%). P0 trigger breakdown: EIH 610 (19.0%), VO_2peak<50%pred 133 (4.1%), peak SBP>220 mmHg 146 (4.6%) (triggers may overlap).

Note (EIH-Red Axiomatic Constraint): The near-100% Red proportions in HTN-EIH (99.3%) and EIH-Only (97.3%) groups are an axiomatic consequence of the P1 label

definition (eih_status = True is an independent sufficient condition for Red), not a model prediction result (see Section 2.4). Model predictions are meaningful only for non-EIH patients; the P1 zone for EIH+ patients is predetermined by the label construction rule. The few EIH+ cases not assigned Red (0 in HTN-EIH; 9 in EIH-Only) had missing eih_status values (NaN does not trigger the EIH-Red condition).

3.5 P0 Pre-exercise Risk Prediction

P0 model performance is summarized in Table 5.

Table 5: P0 pre-exercise risk model comparison (Table 4)

Model	Variant	AUC-ROC	AUPRC	Brier	CV-AUC (mean±SD)
LASSO	with BP	0.582	0.284	0.174	0.622±0.030
LASSO	no BP	0.582	0.284	0.174	0.622±0.030
XGBoost	with BP	0.583	0.287	0.180	0.623±0.028
XGBoost	**no BP**	**0.583**	**0.287**	**0.180**	**0.623±0.028**

LeakageGuard validated (pass). Resting BP data were absent in all records; with-BP and no-BP variants yield identical results. P0 AUC≈0.58 is attributable to absence of medication data (30 fields, all zero), missing resting hemodynamics, and LVEF.

The ROC curve is shown in Figure 3. SHAP analysis (Figure 4) identified age and HTN history as the most important predictors.

3.6 P1 Post-exercise Safety Zone Classification

P1 model performance is summarized in Table 6.

Table 6: P1 post-exercise safety zone classification (Table 5)

Model	Sample	$F1_{macro}$	κ	Accuracy	CV-F1 (mean±SD)
Ordinal Logistic	full	0.386	0.225	0.504	0.393±0.013
LightGBM	full	0.465	0.258	0.499	0.460±0.007
CatBoost	**full**	**0.499**	**0.302**	**0.528**	**0.447±0.015**

Cycling-only vs. full-sample results were identical (exercise_protocol field entirely unknown). Best model: CatBoost with $F1_{macro}$=0.499, Cohen's κ=0.302.

Confusion matrix analysis (Figure 5, CatBoost): Green recall 67.1% (173/258), Yellow recall 51.8% (114/220), Red recall 30.8% (49/159).

SHAP analysis (Figure 6) identified VO_2peak and HR_{peak} as the most important classification features.

Cost-sensitive sensitivity analysis (Table 7):

Table 7: Cost-sensitive vs. standard model comparison (Table 6)

Model	$F1_{macro}$	κ	Green Recall	Yellow Recall	Red Recall
LightGBM (standard)	0.465	0.258	—	—	23.9% (38/159)
LightGBM (CS, Red×4)	**0.392**	**0.128**	—	—	**53.5% (85/159)**
CatBoost (standard)	0.499	0.302	67.1%	51.8%	30.8%
CatBoost (CS)	0.187	0.023	—	—	96.2% (collapsed)

CS = cost-sensitive; Red×4 = Red class weight 4×. LightGBM CS trades F1 (−0.07) for Red recall (+29.6%); CatBoost CS collapses to predicting Red for all.

3.7 P1 Proxy Leakage Ablation

To quantify the proxy leakage contribution of `vo2_peak` (absolute value) toward reconstructing the `vo2_peak_pct_pred` threshold in P1 zone labels, a three-variant ablation study was conducted (Table ??).

Table 8: P1 proxy leakage ablation study (Table 8)

Variant	Feature set	$F1_{macro}$	κ	Green Rec.	Yellow Rec.	Red Rec.
Full	All P1 features (incl. vo2_peak)	0.458	0.237	0.630	0.481	0.268
No-VO2	Exclude vo2_peak	0.455	0.243	0.627	0.481	0.261
VO2-only	Only vo2_peak + age + sex	0.441	0.200	0.612	0.476	0.242

Full vs. No-VO2: ΔF1=+0.003 (negligible); VO2-only (leakage upper bound) F1=0.441.

The Full vs. No-VO2 F1 difference (0.003) is negligible, indicating that `vo2_peak` absolute values have limited capacity to proxy the label threshold in this dataset. The primary reason is the low predictive power of the reference equation (R^2=0.298), which limits the model's ability to reconstruct `vo2_peak_pct_pred` from `vo2_peak` + age + sex. The leakage upper bound (VO2-only, F1=0.441) also falls below Full (F1=0.458), confirming that the P1 F1≈0.50 ceiling reflects inherent feature set limitations rather than leakage-driven inflation.

3.8 EIH Independent Predictors

Multivariable Logistic regression (Table 8, Figure 7) identified HTN history (aOR=1.441, p<0.001), BMI (aOR=1.585, p<0.001), and female sex (aOR=0.862, p=0.005) as independent EIH predictors. The positive association of VO_2peak with EIH after adjustment (aOR=1.230, p<0.001) may reflect exercise intensity selection bias.

Table 9: EIH independent predictors — Logistic regression (Table 7)

Variable	Univariable		Multivariable	
	OR (95% CI)	p	aOR (95% CI)	p
Age	0.942 (0.865–1.028)	0.185	0.975 (0.896–1.062)	0.590
Female sex	1.003 (0.916–1.091)	0.949	0.862 (0.766–0.953)	0.005
VO_2peak	0.979 (0.880–1.064)	0.658	1.230 (1.109–1.346)	<0.001
HTN history	1.410 (1.306–1.532)	<0.001	1.441 (1.315–1.606)	<0.001
BMI	1.603 (1.399–1.828)	<0.001	1.585 (1.366–1.843)	<0.001
HTN duration (years)	1.132 (1.049–1.236)	0.003	0.930 (0.832–1.012)	0.154

N=3,206, EIH positive rate=19.0% (610/3,206); multivariable model converged.

4 Discussion

4.1 Main Findings

This study provides the first large-scale systematic characterization of CPET safety zone distributions in elderly hypertensive patients using a HTN×EIH 2×2 factorial design (N=3,206). Three key findings emerge:

First, HTN and EIH affect distinct cardiopulmonary dimensions through independent main effects. HTN history demonstrated a medium-sized negative effect on aerobic capacity (VO_2peak η^2=0.084) and chronotropic response (HR_{peak} η^2=0.133). EIH, by contrast, primarily manifested as a large elevation in peak blood pressure (η^2=0.371), with negligible independent effect on VO_2peak. Crucially, the VO_2peak interaction was non-significant (F=0.02, p=0.894, η^2=0.000). Although VO_2peak%pred (p=0.025), HR_{peak} (p<0.001), and peak SBP (p=0.023) reached statistical significance for interaction terms, all interaction η^2 values were negligible (≤0.004), indicating no clinically meaningful synergistic impairment. This corrects the intuitive assumption of additive aerobic deterioration in HTN-EIH patients [17]. The large sample (N=3,206) amplifies detection of trivial effects; effect size (η^2) rather than p-values is the appropriate interpretation criterion.

Second, the summary-level predictive ceiling is clearly quantified. P0 AUC≈0.58 and P1 F1≈0.50 represent the approximate ceiling achievable from CPET summary-level data. The primary constraints are: (a) complete absence of medication data (β-blockers critically affect CPET interpretation); (b) missing resting hemodynamics (BP, HR, LVEF); (c) the structural EIH-Red axiom (EIH+ patients are by definition Red, yet eih_status must be excluded to prevent leakage, creating an irreducible feature-label information gap).

Third, the Leakage Guard framework provides a reproducible benchmark. The explicit quantification of leakage effect (≈0.37 F1 units: leaked F1=0.84 vs. clean F1=0.47) provides a concrete reference for retrospective CPET modeling. This gap underscores the importance of systematic leakage audit, particularly for CPET data where test-derived variables often participate in outcome definitions.

4.2 Comparison with Literature

Our P0 AUC=0.583 is lower than Li et al. [9] (AUC=0.794 in CAD patients) and Arina et al. [10] (AUC=0.946 in perioperative elderly patients). These differences are expected given feature richness: both studies included medications, echocardiographic parameters (LVEF), and laboratory biomarkers—categories entirely absent in our dataset. Rather than a methodological failure, our result defines the performance envelope of pure summary-level CPET data without ancillary clinical information.

Hearn et al. [11] and Zignoli et al. [12] demonstrated that deep learning applied to breath-by-breath CPET waveforms substantially improves ventilatory threshold detection accuracy, supporting our Stage II plan to incorporate raw breath-by-breath data.

Our safety zone definition is broadly consistent with the Chinese CPET expert consensus [13] in using VO_2peak %predicted and VE/VCO_2 slope as primary stratification variables. Our data provide large-scale evidence supporting the consensus recommendation that EIH should be considered a high-risk indicator (99.3% Red rate in HTN-EIH; 97.3% in EIH-Only).

4.3 Methodological Contributions

The **dual-timeline P0/P1 design** formally separates pre-exercise risk assessment (P0, using only pre-test variables) from post-exercise stratification (P1, using full CPET results). This design, enforced at the code level through Leakage Guard, ensures that P0 models can be applied in a clinically meaningful way: to screen patients before CPET and identify those requiring closer monitoring.

The **Leakage Guard methodology** addresses a systematic gap in retrospective CPET modeling [8]: the coincidence of EIH classification in both label definition and potential features. By pre-specifying excluded fields and reporting the leakage effect size, we provide a framework reproducible in future studies.

4.4 Clinical Utility and Model Positioning

P0 clinical utility (Table **??**): Surprisingly, a 2-variable baseline model (HTN history + BMI) achieved AUC=0.611, outperforming XGBoost with all features (AUC=0.577,

ΔAUC=-0.034). This finding reinforces the "summary-level predictive ceiling" narrative: complex ML models offer no incremental discriminative value over simple clinical rules when medication, hemodynamic, and laboratory features are absent. Despite this, P0 demonstrates clinical utility as a **rule-out tool**: at the optimal threshold (0.141), NPV=0.853 (when the model predicts low-risk, 85.3% are truly low-risk), with a net benefit advantage over treat-all strategy at both 10% and 20% decision thresholds.

Table 10: P0 clinical utility summary (Table 9, optimal threshold 0.141)

Model	AUC	NPV	NNS	Net Benefit ($\geq$ Treat-All)
2-var baseline (HTN + BMI)	0.611	—	—	—
XGBoost (all features)	0.577	0.853	3.8	Yes (at 10%–20% thresholds)

NPV=Negative Predictive Value; NNS=Number Needed to Screen per true positive.

P1 clinical utility: Red zone NPV=0.781 (when model predicts "non-Red", 78.1% are truly non-Red), but dangerous Red$\rightarrow$Green misclassification rate is 29.4% (45/153 true-Red patients predicted as Green). Notably, a simple VO_2peak threshold rule (<20.1 mL/kg/min) achieves higher Red recall (48.4%) than LightGBM (26.8%), consistent with the predictive ceiling: in the high-Red-recall clinical scenario, a simple threshold may outperform complex models constrained by summary-level features.

Based on the above analysis, the clinical value of the P0/P1 models resides in: (1) **Methodological framework validation**—the HTN$\times$EIH 2$\times$2 design with Leakage Guard provides a reproducible analytical paradigm; (2) **Feature importance discovery**—SHAP results identify VO_2peak and HR_{peak} as primary drivers, informing Stage II feature selection; (3) **Baseline benchmark**—current performance (P0 AUC=0.583, P1 F1=0.499) quantifies the summary-level ceiling against which Stage II/III improvements can be measured. **Direct clinical deployment is not recommended** at current performance levels.

4.5 Limitations

(1) Single-center retrospective design with potential selection bias; no external validation cohort. (2) Medication data entirely absent (30 fields all zero), including β-blockers that critically affect heart rate and CPET interpretation. (3) EIH defined from group code derivation rather than continuous SpO_2 monitoring, introducing classification error. (4) Summary-level CPET data only; breath-by-breath waveforms not available. (5) The EIH+ 100% Red proportion is a structural axiom of the P1 label definition, not a model prediction; subgroup performance in EIH+ patients does not reflect real predictive ability.

(6) Reference equation VO_2peak R^2=0.298 suggests high individual heterogeneity in the elderly population; %pred thresholds may require population-specific calibration.

4.6 Future Directions

Stage II will incorporate breath-by-breath gas exchange and 12-lead ECG waveforms, expected to surpass the summary-level ceiling [11, 12]. Stage III will explore whether wearable devices (e.g., Apple Watch) can serve as home proxies for CPET anchors [18], enabling a home exercise safety corridor (C_{home}). Prospective multicenter validation incorporating medication records and resting hemodynamics is planned.

5 Conclusions

We established the first CPET-based exercise safety zone stratification framework for elderly hypertensive patients using an HTN×EIH 2×2 design (N=3,206). Key conclusions: (1) HTN and EIH affect distinct cardiopulmonary dimensions through independent main effects—HTN reduces aerobic reserve while EIH amplifies blood pressure stress response; the HTN×EIH interaction on VO_2peak is non-significant; (2) Summary-level CPET data approaches its predictive ceiling (P0 AUC=0.583, P1 F1=0.499), establishing the need for breath-by-breath waveform data in Stage II; (3) The Leakage Guard mechanism quantified the data leakage effect at ≈0.37 F1 units, providing a reproducible methodology for preventing data leakage in retrospective CPET modeling. This framework establishes laboratory anchor points for subsequent breath-by-breath modeling and home exercise safety corridor research.

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Figure Captions

Figure 1: HTN×EIH two-way interaction plots (VO_2peak group means). CTRL vs. HTN groups differ significantly (η^2=0.084, medium effect); EIH+/EIH− groups within the same HTN stratum show similar VO_2peak, with non-significant interaction (p=0.894).

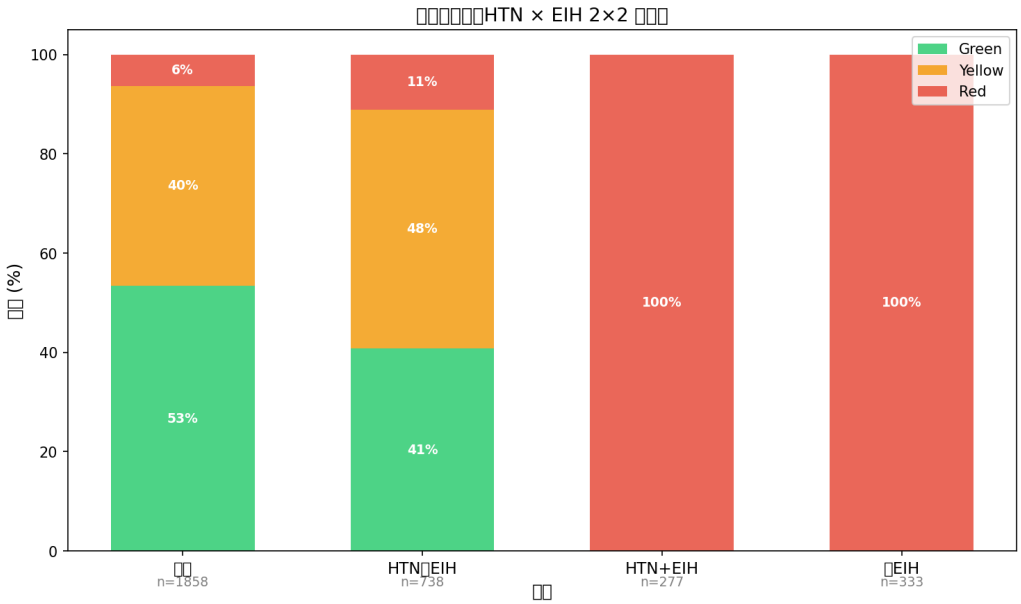

Figure 2: Safety zone distribution (Green/Yellow/Red) by HTN×EIH 2×2 cohort (stacked bar chart). EIH+ groups (HTN-EIH and EIH-Only) are near-entirely Red (99.3%/97.3%), reflecting the EIH-Red axiom in P1 label definition.

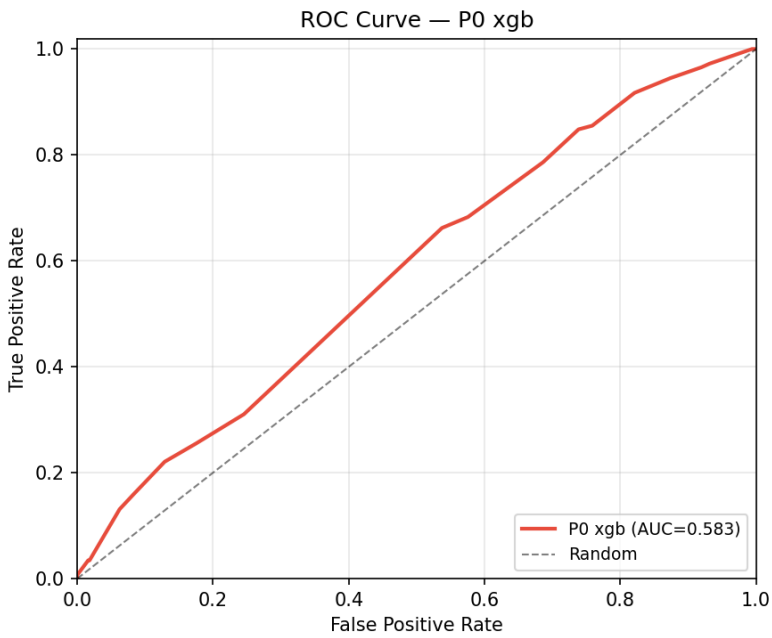

Figure 3: P0 XGBoost ROC curve (no-BP variant, AUC=0.583). The curve near the diagonal reflects the predictive limitation of summary-level data.

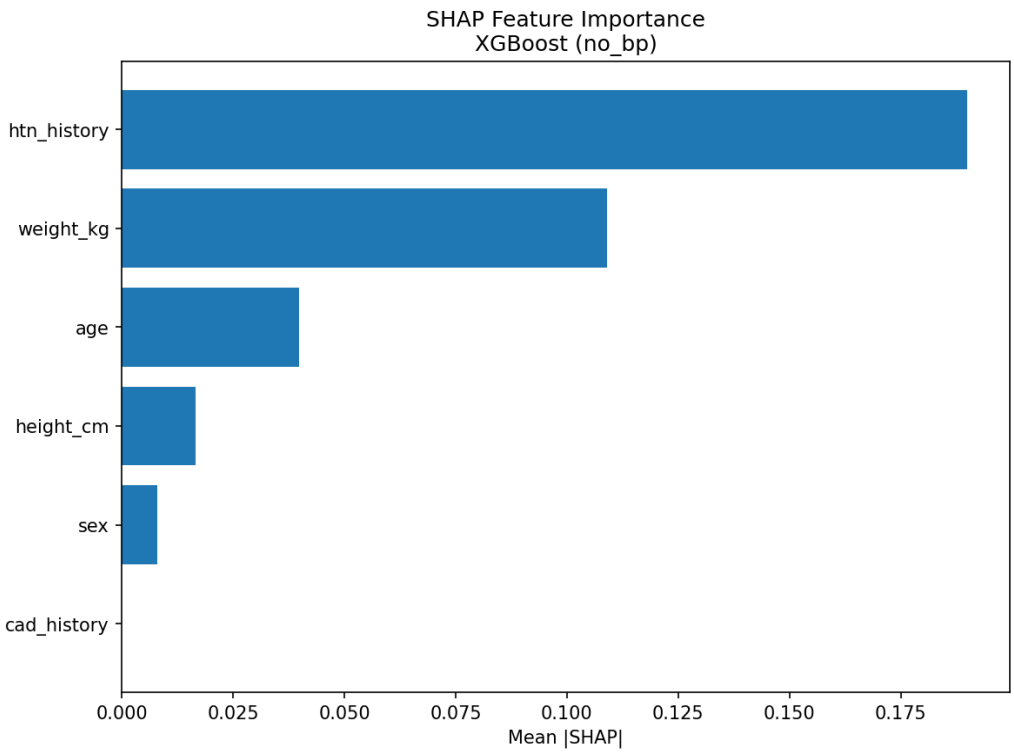

Figure 4: P0 XGBoost SHAP global feature importance (no-BP variant). Age and HTN history are the most important predictors.

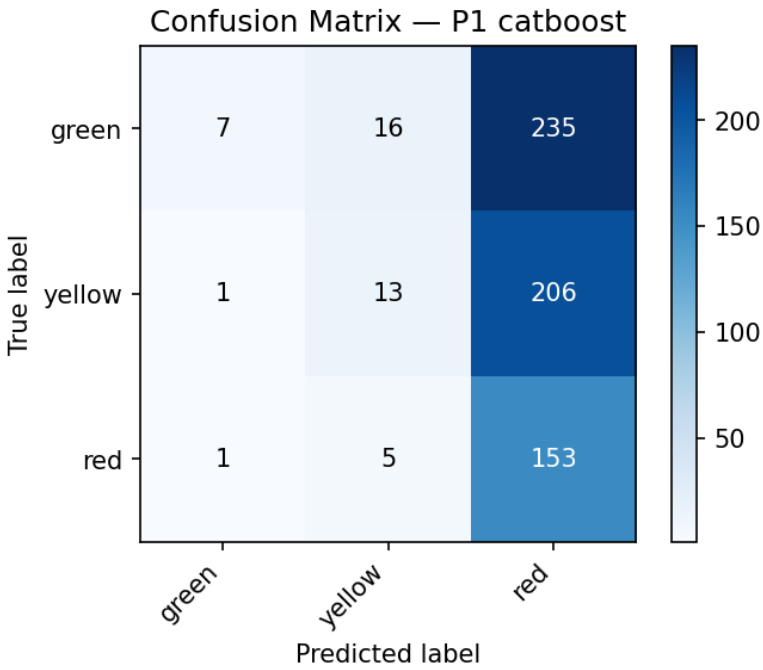

Figure 5: P1 CatBoost best-model confusion matrix ($F1_{macro}$=0.499, κ=0.302). Green recall 67.1% (173/258); Red recall 30.8% (49/159).

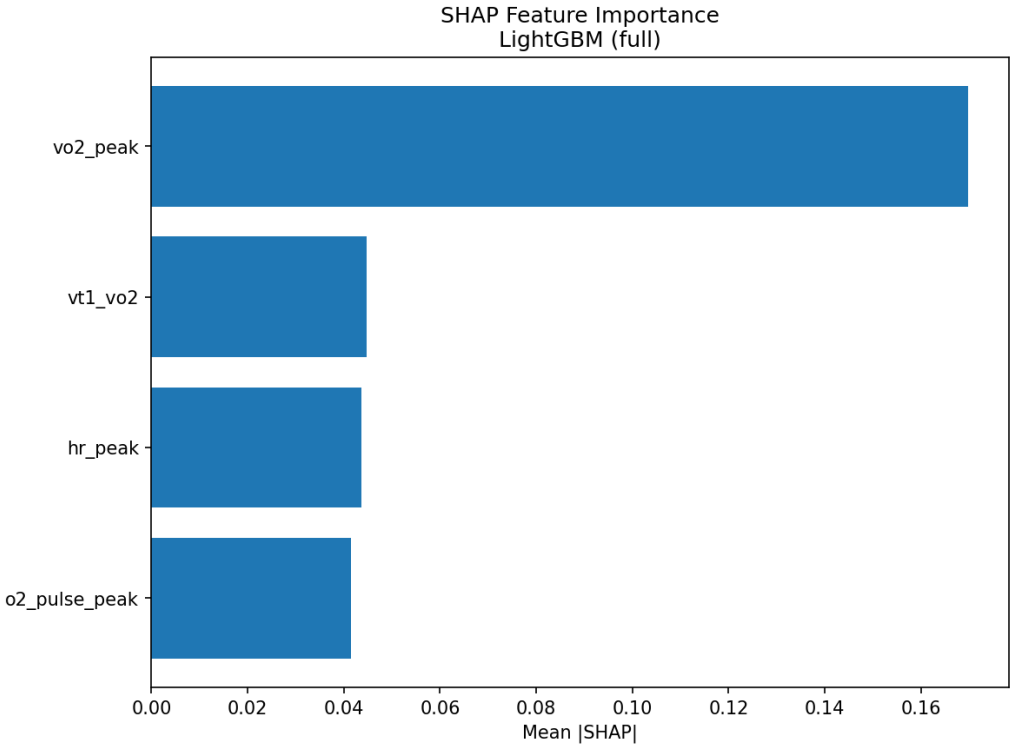

Figure 6: P1 LightGBM SHAP global feature importance (full-sample variant). VO_2peak and peak HR are the dominant classification features.

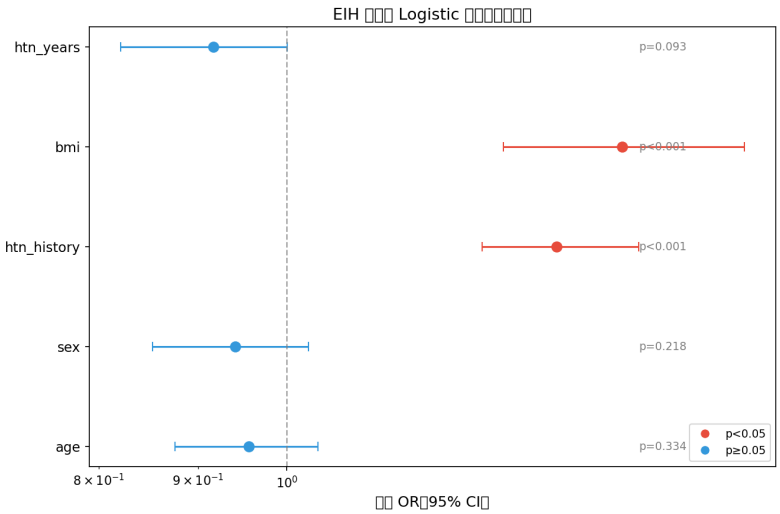

Figure 7: Forest plot of EIH independent predictors (multivariable Logistic regression). HTN history (aOR=1.44) and BMI (aOR=1.59) are the strongest independent predictors.